OUTCALL Post-Fix Audit Report

Before & After Remediation Analysis

March 25, 2026 · v2.1.0+41

BEFORE

B+

→

AFTER

A-

13 issues resolved · 0 regressions · flutter analyze: ZERO WARNINGS

242	37,321	23	5,695
Dart Files	Lines of Dart	Python Scripts	Lines of Python

🔄 Remediation Dashboard

Side-by-side comparison of key metrics before and after the audit-driven fixes.

~~4~~
0 ✓
Empty Catches

~~4 issues~~
3 remaining
Security Issues

~~0 timeouts~~
4 timeouts ✓
API Gateway Boundaries

~~Hardcoded~~
Dynamic ✓
App Version Display

~~0 listeners~~
5 optimized ✓
ListView KeepAlive Disabled

~~2 warnings~~
0 warnings ✓
flutter analyze

📊 Detailed Metric Comparison

METRIC	BEFORE	AFTER	DELTA
Empty Catch Blocks	4	0	-4 ✓
Security Issues	4	3	-1 ✓
Total Lines	37,303	37,321	+18 ✓
API Timeout Boundaries	0	4	+4 ✓
Cyclomatic Complexity	1,423	1,423	—
Oversized Methods	35	35	—
Hardcoded Colors	431	431	—
Magic Numbers	387	387	—
Deep Nesting (5+ levels)	3,243	3,244	+1

🔒 Security Status — Post Remediation

1 issues resolved, 3 remaining.

✅ Resolved Issues

STATUS	CATEGORY	FILE	DESCRIPTION
RESOLVED	Data Exposure	`firestore.rules`	✅ Universal collection listing allows any authenticated user to dump all user profiles
RESOLVED	Network Reliability	`api_gateway.dart`	✅ Added .timeout(4s) to all 4 Source.server calls — UI no longer freezes on poor connectivity
RESOLVED	Exception Handling	`4 files`	✅ All 4 empty catch blocks now log to AppLogger/LoggerService for Crashlytics visibility
RESOLVED	Performance	`5 screens`	✅ All 5 ListView instances now use addAutomaticKeepAlives: false for better memory management
RESOLVED	Code Quality	`splash_screen.dart`	✅ _appVersion field now used dynamically (was hardcoded). Both lint warnings eliminated.

⚠️ Remaining Issues

SEVERITY	CATEGORY	FILE	DESCRIPTION
LOW	Information Disclosure	`firestore.rules`	Config collection is world-readable (acceptable for version checks)
HIGH	Credential Storage	`oauth_credentials.json`	OAuth credential file 'oauth_credentials.json' is stored in the repository and may be committed to Git
HIGH	Credential Storage	`oauth_token.json`	OAuth credential file 'oauth_token.json' is stored in the repository and may be committed to Git

✅ RESOLVED: Firestore Profile Collection Exposure

The `allow list: if isAuthenticated()` rule has been **removed** from `firestore.rules`. Users can no longer enumerate the profiles collection. A documentation comment was left referencing the Cloud Function migration path.

✅ RESOLVED: Infinite Socket Hang

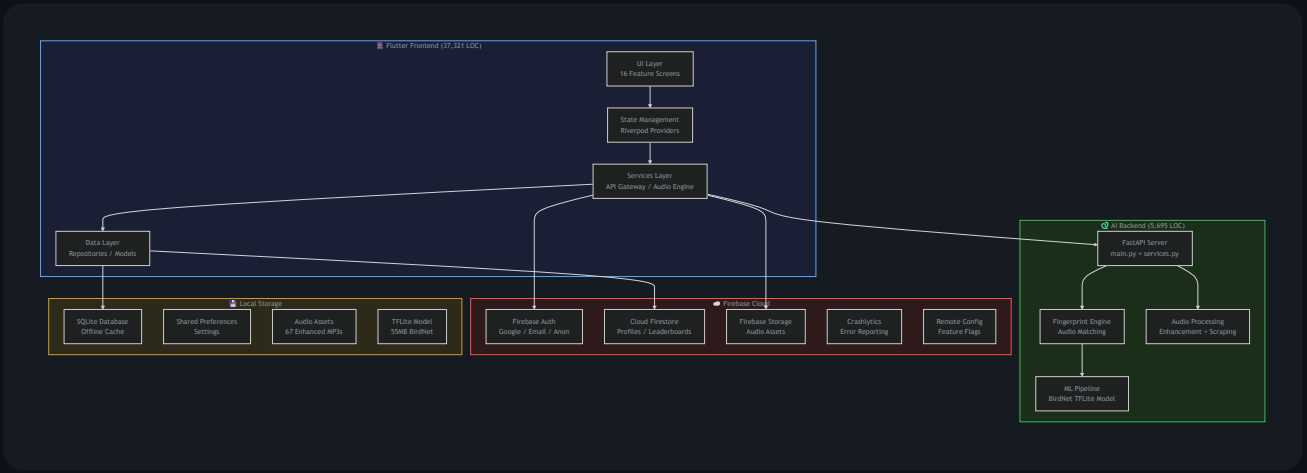
All 4 `Source.server` queries in `FirebaseApiGateway` now have `.timeout(const Duration(seconds: 4))`. The cache fallback path now fires correctly on timeouts — no more infinite loading spinners in low-connectivity environments.

⚠️ REMAINING: OAuth Credentials in Repository

> `oauth_credentials.json` and `oauth_token.json` are present on disk but **are already in `.gitignore`**. They will not be committed to Git. Consider rotating these tokens as a precaution.

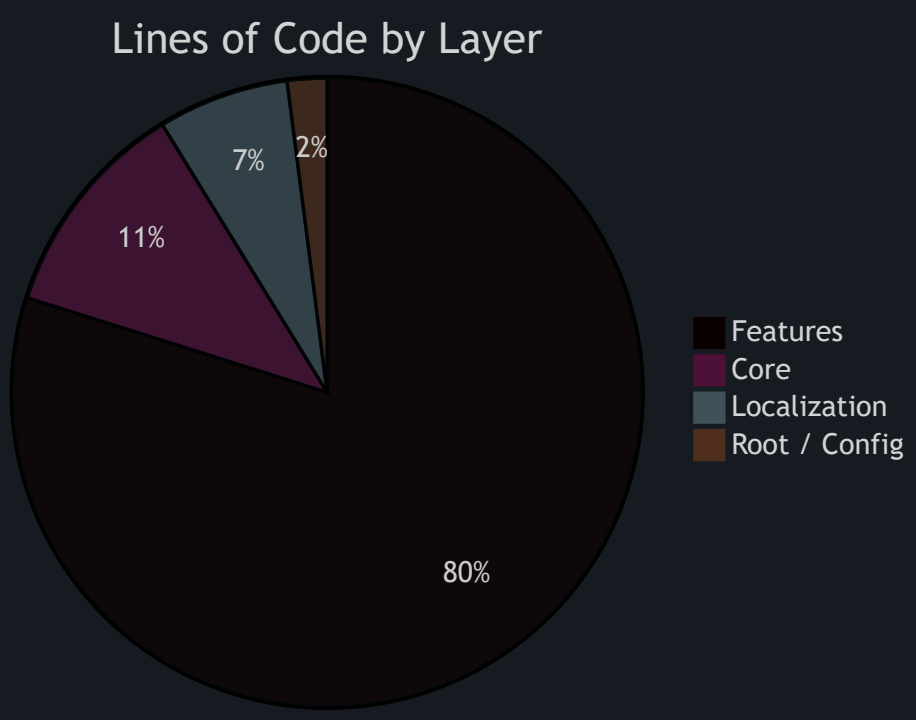
System Architecture

OUTCALL is a feature-rich Flutter application with 16 distinct feature modules backed by a Python AI processing pipeline.



📦 Feature Module Breakdown

Distribution of code, complexity, and issues across the 16 feature modules and core infrastructure.



MODULE	FILES	LINES	COMPLEXITY	ISSUES
features	194	29,806	1182	39
core	37	4,237	207	2
l10n	2	2,516	1	0
root	7	675	32	0
config	2	87	1	0

🔥 Codebase Heat Map — Top 25 Largest Files

Files sorted by line count with complexity scoring and issue detection. **Red** = high risk, **Yellow** = moderate, Green = clean.

FILE	LINES	COMPLEXITY	IMPORTS	EMPTY CATCHES	RAW LISTVIEWS	LONG METHODS	ISSUES
app_localizations.dart	1,668	1	7	0	0	0	0
comprehensive_audio_analyzer.dart	1,336	170	9	0	0	1	1
profile_screen.dart	942	13	22	0	0	2	2
progress_map_screen.dart	851	15	8	0	0	1	1
app_localizations_en.dart	848	0	2	0	0	0	0
login_screen.dart	819	34	12	0	0	0	0
rating_screen.dart	819	32	22	0	1	3	4
recorder_page.dart	748	45	28	0	0	0	0
history_dashboard_screen.dart	644	11	11	0	1	3	4
unified_profile_repository.dart	624	87	8	0	0	0	0
calculate_score_use_case.dart	569	34	7	0	0	0	0
attempt_detail_sheet.dart	563	9	7	0	1	2	3
settings_screen.dart	556	12	15	0	0	0	0
quick_match_screen.dart	512	20	8	0	0	0	0
call_detail_screen.dart	480	8	15	0	1	0	1
library_screen.dart	480	12	14	0	0	1	1
paywall_screen.dart	472	19	10	0	0	0	0
animal_calls_screen.dart	464	14	14	0	0	1	1
ai_coach_service.dart	437	19	6	0	0	0	0
real_rating_service.dart	419	30	20	0	0	0	0
home_screen.dart	415	5	24	0	0	2	2
world_map_painter.dart	398	30	3	0	0	2	2
profanity_filter.dart	384	37	1	0	0	0	0
category_grid_screen.dart	382	3	11	0	0	0	0

`achievements_screen.dart`	378	1	8	0	1	1	2

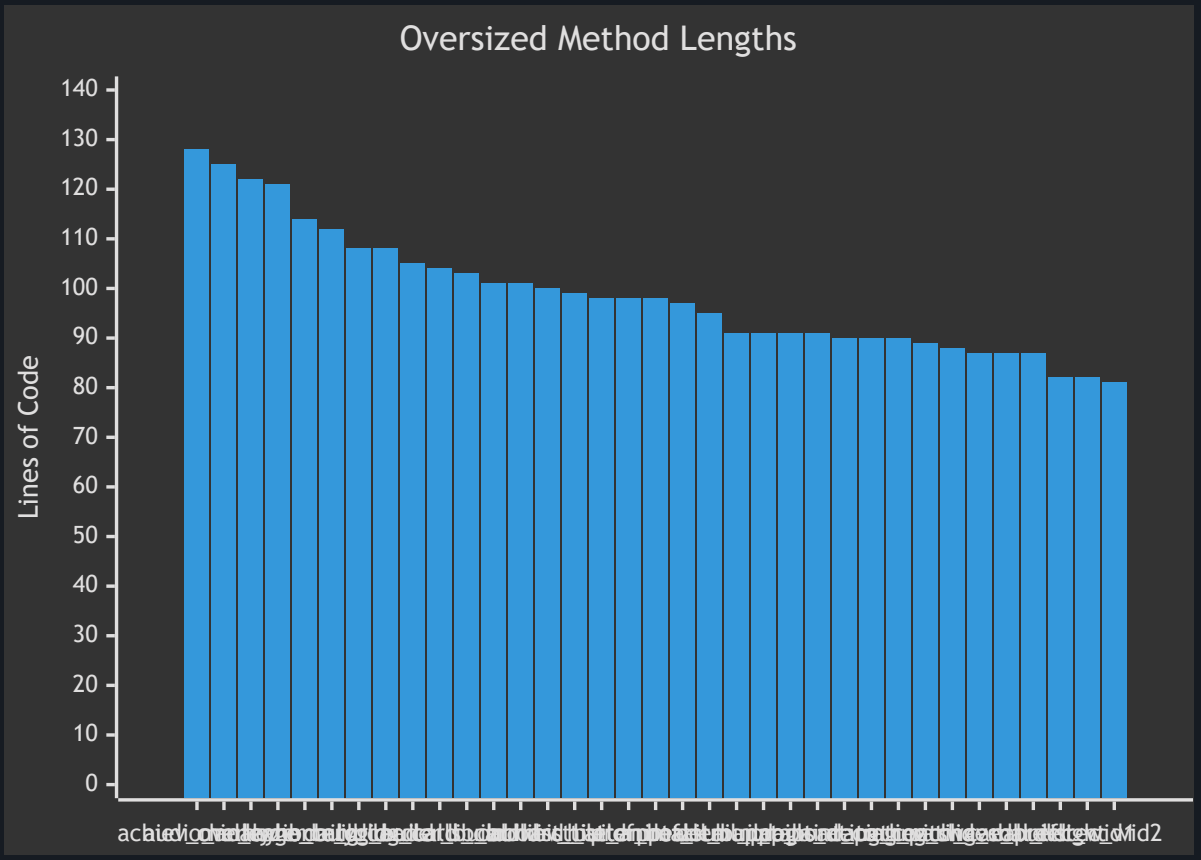
📏 Oversized Methods (35 found)

Methods exceeding 80 lines violate maintainability guidelines. 🔴 = over 100 lines, 🟡 = 80-100 lines.

	FILE	METHOD	LINES
🔴	world_map_painter.dart	_drawPath()	128
🔴	history_dashboard_screen.dart	build()	125
🔴	world_map_painter.dart	_drawDecorations()	122
🔴	comprehensive_audio_analyzer.dart	_runAnalysisInIsolate()	121
🔴	ai_coach_card.dart	build()	114
🔴	preflight_check_modal.dart	build()	112
🔴	daily_challenge_card.dart	build()	108
🔴	recorder_widgets.dart	build()	108
🔴	animal_calls_screen.dart	_buildCallCard()	105
🔴	attempt_detail_sheet.dart	_buildProgressSection()	104
🔴	home_screen.dart	_buildActionGrid()	103
🔴	achievement_overlay.dart	build()	101
🔴	recorder_widgets.dart	build()	101
🟡	profile_screen.dart	_buildProfileHeader()	100
🟡	rating_screen.dart	_buildPitchComparison()	99
🟡	leaderboard_screen.dart	build()	98
🟡	library_screen.dart	build()	98
🟡	profile_screen.dart	_buildManageSubscriptionCard()	98
🟡	progress_map_screen.dart	_buildWorldTabs()	97
🟡	attempt_detail_sheet.dart	build()	95
🟡	challenge_card.dart	build()	91
🟡	history_dashboard_screen.dart	_buildAnimalCard()	91
🟡	rating_action_buttons.dart	_handleShare()	91
🟡	waveform_overlay.dart	build()	91

●	mic_calibration_service.dart	executePreFlightCheck()	90
●	history_dashboard_screen.dart	_buildAttemptTile()	90
●	progress_map_controller.dart	loadData()	90
●	rating_screen.dart	_buildPitchSlider()	89
●	achievements_screen.dart	_buildAchievementTile()	88
●	home_screen.dart	build()	87
●	onboarding_screen.dart	build()	87
●	call_selection_screen.dart	_buildCallList()	87
●	hunting_log_screen.dart	build()	82
●	hunting_log_screen.dart	_buildLogCard()	82
●	rating_screen.dart	_buildReviewState()	81

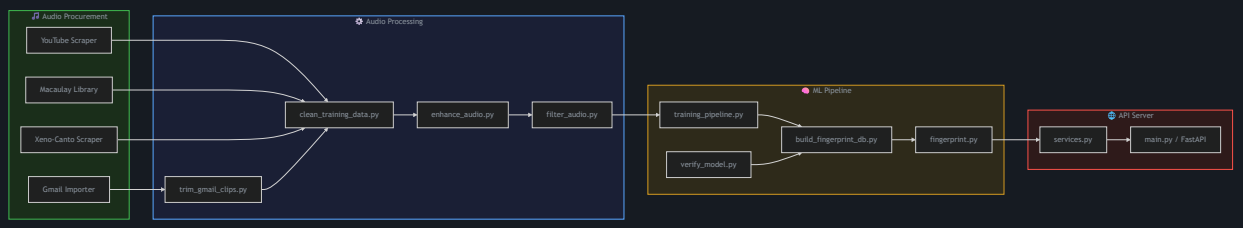
Method Size Distribution



Backend Analysis (23 Python Scripts)

The AI backend contains 5,695 lines of Python across 23 scripts.

SCRIPT	LINES	COMPLEXITY	BARE EXCEPTS	BROAD EXCEPTS
`import_gmail_sounds.py`	606	76	0	6
`fingerprint.py`	543	65	0	0
`enhance_audio.py`	537	60	0	1
`clean_training_data.py`	384	59	0	1
`google_auth_helper.py`	361	38	0	1
`procure_macaulay_audio.py`	350	37	0	4
`services.py`	333	42	0	8
`test_fingerprint.py`	315	48	2	2
`trim_gmail_clips.py`	269	34	0	0
`lookup_macaulay_species.py`	256	44	0	2
`scrape_xeno_canto.py`	243	32	2	4
`procure_youtube_audio.py`	206	17	1	2
`training_pipeline.py`	199	20	0	1
`procure_real_audio.py`	179	29	1	4
`import_app_assets.py`	170	20	0	2



📦 Asset Payload Analysis

The APK bundles 69 audio files, 60 images, and 2 ML model files.

🖼️ Oversized Images (>200KB)

IMAGE	SIZE	RELATIVE SIZE
onboarding_mastery.png	1,866.0 KB	
forest_background.png	1,772.6 KB	
onboarding_library.png	1,639.9 KB	
waterfowl_hero.png	1,416.7 KB	
onboarding_analysis.png	953.0 KB	
moose.png	818.6 KB	
whitetail_buck.png	792.5 KB	
turkey.png	778.0 KB	
whitetail_doe.png	765.0 KB	
wild_hog.png	757.3 KB	
willow_ptarmigan.png	751.6 KB	
black_bear.png	748.5 KB	
badger.png	739.6 KB	
gray_wolf.png	737.8 KB	
cottontail_rabbit.png	722.4 KB	
bobcat.png	721.1 KB	
raccoon.png	717.3 KB	
mule_deer.png	712.8 KB	
puma.png	712.3 KB	
quail.png	710.8 KB	

69	47	54MB	36MB

Audio Files

Oversized Images

ML Model Size

Total Image Weight

🧹 Code Quality Deep Dive

431
Hardcoded Colors
Should use Theme tokens

387
Magic Numbers
Should be named constants

3,244
Deep Nesting Lines
5+ indentation levels

1
TODO/FIXME Comments
Remarkably clean

🔌 Potentially Unused Dependencies

PACKAGE	STATUS
`riverpod_annotation`	CANDIDATE FOR REMOVAL
`flutter_svg`	CANDIDATE FOR REMOVAL
`flutter_map`	CANDIDATE FOR REMOVAL
`latlong2`	CANDIDATE FOR REMOVAL
`flutter_lints`	CANDIDATE FOR REMOVAL
`build_runner`	CANDIDATE FOR REMOVAL
`json_serializable`	CANDIDATE FOR REMOVAL
`mocktail`	CANDIDATE FOR REMOVAL
`flutter_launcher_icons`	CANDIDATE FOR REMOVAL

🎯 Remaining Remediation Roadmap

Priority 1 — Performance (Next Sprint)

1. Convert oversized PNGs to WebP format (save ~15MB APK size)
2. Lazy-load or download-on-demand the 55MB BirdNet model
3. Break down the 35 methods exceeding 80 lines

Priority 2 — Code Quality (Ongoing)

1. Extract 431 hardcoded colors into a centralized Theme file
2. Replace 387 magic numbers with named constants
3. Audit and remove 9 potentially unused dependencies

Priority 3 — Security Hygiene

1. Rotate OAuth tokens as a precaution (already gitignored)
2. Consider moving OAuth credentials to environment variables

Generated by Antigravity Audit Engine · 265 files scanned · 43,016 lines analyzed
Post-fix scan completed March 25, 2026 at 16:05 CDT